

Chapter 5. Factor groups

5.2 Factor groups

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ \pi \downarrow & \nearrow \tilde{f} & \\ G / \ker(f) & & \end{array}$$

In this video, we will prove some properties of factor groups.

Some properties of factor groups

THEOREM 5.9. *Let H be a normal subgroup of a group G .*

(1) *If G is abelian, then G/H is abelian.*

(2) *If $G = \langle g \rangle$ is cyclic, then G/H is also cyclic: $G/H = \langle Hg \rangle$.*

Proof

① Suppose that G is abelian. For any $a, b \in G$ we need to show that

$$Ha \cdot Hb \stackrel{?}{=} Hb \cdot Ha$$

$$\underline{Ha \cdot Hb = H(ab) = H(ba) = Hb \cdot Ha}$$

② Suppose that $G = \langle g \rangle$ and let us show that $G/H \stackrel{?}{=} \langle Hg \rangle$. To do so, we need to check that every element of G/H is a power of Hg . Take $a \in G$ and consider the coset $\underline{Ha \in G/H}$.

$$a \in G = \langle g \rangle \Rightarrow a = g^k \text{ for some integer } k.$$

$$\underline{Ha = Hg^k = (Hg)^k \in \langle Hg \rangle}.$$

And so, $G/H = \langle Hg \rangle$ ■

Example. Prove $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}_6$.

$\mathbb{Z}$ cyclic $\Rightarrow \mathbb{Z}/6\mathbb{Z}$ is cyclic.

We know that $\mathbb{Z}/6\mathbb{Z}$ is a cyclic group of 6 elements. So,

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}_6$$
